

Lecture 4: Standard Errors

Wednesday 2nd September, 2026

Today's Plan

- ▶ Recap OLS and various forms of standard errors
- ▶ Standard errors are tedious but I guess you are supposed to know this stuff
- ▶ Hopefully first and last time we talk about this

Recap: Asymptotics for OLS and the Linear Model

$$y_i = \beta_0 + \beta x_i + u_i$$

Recall the three basic OLS assumptions

1. $\mathbb{E}(u_i | X_i) = 0$
2. $(X_i, Y_i), i = 1, \dots, n$, are i.i.d.
3. Large outliers are rare $\mathbb{E}[Y^4] < \infty$ and $\mathbb{E}[X^4] < \infty$.

Unbiasedness and Consistency

- Unbiasedness means on average we don't over or under estimate $\hat{\beta}$

$$\mathbb{E}[\hat{\beta}] - \beta_0 = 0$$

- Consistency tells us that we approach the true β_0 as $n \rightarrow \infty$.

$$\hat{\beta} \xrightarrow{p} \beta_0$$

- Example: $X_{(1)}$ is unbiased but not consistent for the mean.
- Example $\frac{n}{n-5} \bar{X}$ is consistent but biased for the mean.

Bias Variance Decomposition

We can decompose any estimator into two components

$$\underbrace{\mathbb{E}[(y - \hat{f}(x))^2]}_{MSE} = \underbrace{\left(\mathbb{E}[\hat{f}(x) - f(x)]\right)^2}_{Bias^2} + \underbrace{\mathbb{E}\left[\left(\hat{f}(x) - \mathbb{E}[\hat{f}(x)]\right)^2\right]}_{Variance}$$

- What minimizes MSE?

$$f(x_i) = \mathbb{E}[Y_i \mid X_i]$$

- In general we face a tradeoff between bias and variance.
- In OLS we minimize the variance among unbiased estimators assuming that the true $f(x_i) = X_i\beta$ is linear. (But is it?)

Outliers and Leverage

One way to find outliers is to calculate the leverage of each observation i . We begin with the hat matrix:

$$P = X(X'X)^{-1}X'$$

and consider the diagonal elements which for some reason are labeled h_{ii}

$$h_{ii} = x_i(X'X)^{-1}x_i'$$

This tells us how influential an observation is in our estimate of $\hat{\beta}$.

Particularly important for $\{0, 1\}$ dummy variables with uneven groups.

Leave One Out Regression

- ▶ This is sometimes called the **Jackknife**
- ▶ Sometimes it is helpful to know what would happen if we omitted a single observation i
- ▶ Turns out we don't need to run N regressions

$$\begin{aligned}\hat{\beta}_{-i} &= (X'_{-i}X_{-i})^{-1}X'_{-i}Y_{-i} \\ &= \hat{\beta} - (X'X)^{-1}x_i\tilde{u}_i \quad \text{where } \tilde{u}_i = (1 - h_{ii})^{-1}\hat{u}_i\end{aligned}$$

- ▶ $\tilde{u}_i$ has the interpretation of the **LOO prediction error**.
- ▶ high leverage observations move $\hat{\beta}$ a lot.

You can read more about this in Ch3 of Hansen. [Skip derivation]

Gauss Markov Adds two assumptions:

1. $E(u_i|X_i) = 0$
2. $(X_i, Y_i), i = 1, \dots, n$, are i.i.d.
3. Large outliers are rare $\mathbb{E}[Y^4] < \infty$ and $\mathbb{E}[X^4] < \infty$.
4. $Var(u_i) = \sigma^2$ (homoskedasticity)
5. $u_i \sim N(0, \sigma^2)$ (normal errors)

Under these assumptions you learned that OLS is **BLUE**

Start with the variance of the residuals to form a **diagonal** matrix D :

$$\text{Var}(\mathbf{u}|\mathbf{X}) = \mathbb{E}(\mathbf{u}\mathbf{u}' | \mathbf{X}) = \mathbf{D}$$

$$\mathbf{D} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2) = \begin{pmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_n^2 \end{pmatrix}$$

- ▶ $\mathbf{D}$ is diagonal because $\mathbb{E}[u_i u_j | X] = \mathbb{E}[u_i | x_i] \mathbb{E}[u_j | x_j] = 0$ (independence)
- ▶ The elements of D_i are given by $\mathbb{E}[u_i^2 | X] = \mathbb{E}[u_i^2 | x_i] = \sigma_i^2$.
- ▶ In the **homoskedastic** case $\mathbf{D} = \sigma^2 \mathbf{I}_n$.

Variance of $\hat{\beta}$

A useful identity for linear algebra:

$$\text{Var}(a\mathbf{Z}) = a^2 \text{Var}(\mathbf{Z})$$

$$\text{Var}(A\mathbf{Z}) = A \text{Var}(\mathbf{Z}) A'$$

Recall that $\text{Var}(\mathbf{Y}|\mathbf{X}) = \text{Var}(\mathbf{u}|\mathbf{X})$ and also recall the formula for $\hat{\beta}$:

$$\hat{\beta} = \underbrace{(X'X)^{-1}X'}_A Y = A'Y$$

$$\begin{aligned}\mathbf{V}_{\hat{\beta}} &= \text{Var}(\hat{\beta}|\mathbf{X}) = (X'X)^{-1}X' \text{Var}(Y|\mathbf{X})X(X'X)^{-1} \\ &= (X'X)^{-1}(X'\mathbf{D}X)(X'X)^{-1}\end{aligned}$$

We have that $(X'\mathbf{D}X) = \sum_{i=1}^N x_i x_i' \sigma_i^2$. Under homoskedasticity $\mathbf{D} = \sigma^2 \mathbf{I}_n$ and $\mathbf{V}_{\hat{\beta}} = \sigma^2 (X'X)^{-1}$.

$$\mathbf{D} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2) = \mathbb{E}(u_i u_i' | \mathbf{X}) = \mathbb{E}(\tilde{\mathbf{D}} | \mathbf{X})$$

We can estimate $\hat{\mathbf{v}}_{\hat{\beta}}$ by plugging in $\mathbf{D} \rightarrow \tilde{\mathbf{D}}$:

$$\begin{aligned}\mathbf{v}_{\hat{\beta}} &= (X'X)^{-1}(X'\tilde{\mathbf{D}}X)(X'X)^{-1} \\ &= (X'X)^{-1} \left(\sum_{i=1}^N x_i x_i' u_i^2 \right) (X'X)^{-1}\end{aligned}$$

The expectation shows us this estimator is unbiased:

$$\begin{aligned}\mathbb{E}[\mathbf{v}_{\hat{\beta}} | X] &= (X'X)^{-1} \left(\sum_{i=1}^N x_i x_i' \mathbb{E}[u_i^2 | X] \right) (X'X)^{-1} \\ &= (X'X)^{-1} \left(\sum_{i=1}^N x_i x_i' \sigma_i^2 \right) (X'X)^{-1} = (X'X)^{-1}(X'DX)(X'X)^{-1}\end{aligned}$$

Heteroskedasticity Consistent (HC) Variance Estimates

What we need is a consistent estimator for $\hat{u}_i^2$.

$$\mathbf{v}_{\hat{\beta}}^{HC0} = (X'X)^{-1} \left(\sum_{i=1}^N x_i x_i' \hat{u}_i^2 \right) (X'X)^{-1}$$

$$\mathbf{v}_{\hat{\beta}}^{HC1} = (X'X)^{-1} \left(\sum_{i=1}^N x_i x_i' \hat{u}_i^2 \right) (X'X)^{-1} \cdot \left(\frac{n}{n-k} \right)$$

Could use leave one out variance estimate:

$$\mathbf{v}_{\hat{\beta}}^{HC2} = (X'X)^{-1} \left(\sum_{i=1}^N (1 - h_{ii})^{-1} x_i x_i' \hat{u}_i^2 \right) (X'X)^{-1}$$

$$\mathbf{v}_{\hat{\beta}}^{HC3} = (X'X)^{-1} \left(\sum_{i=1}^N (1 - h_{ii})^{-2} x_i x_i' \hat{u}_i^2 \right) (X'X)^{-1}$$

Heteroskedasticity Consistent (HC) Variance Estimates

- ▶ We know that $\mathbf{v}_{\hat{\beta}}^{HC3} > \mathbf{v}_{\hat{\beta}}^{HC2} > \mathbf{v}_{\hat{\beta}}^{HC0}$ because $(1 - h_{ii}) < 1$.
- ▶ $HC3$ are the most **conservative** and also place the most weight on potential outliers.
- ▶ Stata uses $HC1$ as the default and it is what most people refer to when they say **robust standard errors**.
- ▶ These are often called White (1980) SE's or Eicher-Huber-White SE's.
- ▶ In small sample some evidence that $HC2$ has better **coverage**, (what is that?)

What is Clustering?

Suppose we want to relax our i.i.d. assumption:

- ▶ Each observation i is a **villager** and each group g is a **village**
- ▶ Each observation i is a **student** and each group g is a **class**.
- ▶ Each observation t is a **year** and each entity i is a **state**.
- ▶ Each observation t is a **week** and each entity i is a **shopper**.

We might expect that $\text{Cov}(u_{g1}, u_{g2}, \dots, u_{gN}) \neq 0 \rightarrow$ independence is a bad assumption.

The groups (villages, classrooms, states) are independent of one another, but within each group we can allow for arbitrary correlation.

- ▶ If correlation is within an individual over time we call it **serial correlation** or **autocorrelation**
- ▶ Just like in time-series→ we have fewer effective independent observations in our sample.
- ▶ Asymptotics now about the number of groups $G \rightarrow \infty$ not observations $N \rightarrow \infty$

Begin by stacking up observations in each group $\mathbf{y}_g = [y_{g1}, \dots, y_{gn_g}]$, we can write OLS three ways:

$$y_{ig} = x'_{ig}\beta + u_{ig}$$

$$\mathbf{y}_g = \mathbf{X}_g\beta + \mathbf{u}_g$$

$$\mathbf{Y} = \mathbf{X}\beta + \mathbf{u}$$

All of these are equivalent:

$$\hat{\beta} = \left(\sum_{g=1}^G \sum_{i=1}^{n_g} x'_{ig} x_{ig} \right)^{-1} \left(\sum_{g=1}^G \sum_{i=1}^{n_g} x'_{ig} y_{ig} \right)$$

$$\hat{\beta} = \left(\sum_{g=1}^G \mathbf{x}'_g \mathbf{x}_g \right)^{-1} \left(\sum_{g=1}^G \mathbf{x}'_g \mathbf{y}_g \right)$$

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1} (\mathbf{X}'\mathbf{Y})$$

Clustering (Continued)

The error terms have covariance within each cluster g as:

$$\Sigma_g = \mathbb{E}(\mathbf{u}_g \mathbf{u}_g' | X_g)$$

In order to calculate $\widehat{V}_{\widehat{\beta}}$ we replace the covariance matrix $\mathbf{D}$ with Ω and consider an estimator $\widehat{\Omega}_n$. We exploit **independence across clusters**:

$$\text{var} \left(\left(\sum_{g=1}^G X_g' \mathbf{u}_g \right) | X \right) = \sum_{g=1}^G \text{var} (X_g' u_g | X_g) = \sum_{g=1}^G X_g' \mathbb{E} (u_g u_g' | X_g) X_g = \sum_{g=1}^G X_g' \Sigma_g X_g \equiv \Omega_N$$

And an estimate of the variance:

$$V_{\widehat{\beta}} = \text{var}(\widehat{\beta} | X) = (\mathbf{X}'\mathbf{X})^{-1} \widehat{\Omega}_n (\mathbf{X}'\mathbf{X})^{-1}$$

$$\begin{aligned}\widehat{\Omega}_n &= \sum_{g=1}^G X'_g \widehat{\mathbf{u}}_g \widehat{\mathbf{u}}'_g X_g \\ &= \sum_{g=1}^G \sum_{i=1}^{n_g} \sum_{\ell=1}^{n_g} x_{ig} x'_{\ell g} \widehat{u}_{ig} \widehat{u}_{\ell g} \\ &= \sum_{g=1}^G \left(\sum_{i=1}^{n_g} x_{ig} \widehat{u}_{ig} \right) \left(\sum_{\ell=1}^{n_g} x_{\ell g} \widehat{u}_{\ell g} \right)'\end{aligned}$$

- First line makes explicit: independence over each of G clusters
- Last line easiest for computer

$$\widehat{V}_{\hat{\beta}}^{\text{CR1}} = (X'X)^{-1} \left(\sum_{g=1}^G X'_g \hat{u}_g \hat{u}'_g X_g \right) (X'X)^{-1}$$
$$\widehat{V}_{\hat{\beta}}^{\text{CR3}} = (X'X)^{-1} \left(\sum_{g=1}^G X'_g \tilde{u}_g \tilde{u}'_g X_g \right) (X'X)^{-1}$$

- Can replace $\hat{\mathbf{u}}_g \rightarrow \tilde{\mathbf{u}}_g$ for leave-one out like *HC3* (these are called *CR3*).

How should I cluster my standard errors?

- ▶ Heck if I know.
- ▶ This is very problem specific
- ▶ It matters a lot → standard errors can get orders of magnitude larger.
- ▶ Do you believe across group independence or not? [this is the only thing that matters]
- ▶ If you include **fixed effects** probably you need at least clustering at that level.

Newey West Standard Errors (HAC)

- ▶ In serially correlated data we need to account for $\text{Cov}(u_t, u_{t-1}, \dots) \neq 0$.
- ▶ Clustering is one solution, but we may end up throwing away all of our data.
- ▶ Instead we could estimate the serial correlation.
- ▶ May also want standard errors that are **heteroskedasticity AND autocorrelation consistent** (HAC).
- ▶ Have to select a number of lags L

$$\hat{\Omega}_{n,L}^{HAC} = \sum_{t=1}^T u_t^2 x_t x_t' + \sum_{l=1}^L \sum_{t=l+1}^T w_l u_t u_{t-l} (x_t x_{t-l}' + x_{t-l} x_t')$$
$$w_l = 1 - \frac{l}{L+1}$$

What about β ?

- ▶ All of the estimates above should produce **identical** point estimates
- ▶ We have just been talking about adjusting **standard errors**
- ▶ Should the presence of heteroskedasticity change our estimates of $\hat{\beta}$ as well?

A simple extension is Weighted Least Squares (WLS)

- ▶ Different motivations
- ▶ Suppose we have sampling weights that are not $\frac{1}{n}$ from survey data, etc:
 - If my population is supposed to represent all US residents and my sample is 75% Women...
 - Relax LSA (2) $(X_i, Y_i), i = 1, \dots, n$, are i.i.d.
- ▶ In this case, OLS is still unbiased and consistent, just **inefficient**

Can weight each observation as w_i so that $\sum_{i=1}^N w_i = 1$ instead of $w_i = \frac{1}{N}$.

Can define a diagonal matrix W with entries w_i .

$$\arg \min_{\beta} \sum_{i=1}^N w_i (y_i - X_i \beta)^2 = \arg \min_{\beta} \|W^{1/2} [Y - X\beta]\|$$

Can also consider a transformation of the data

$$\begin{aligned}\tilde{y}_i &= \sqrt{w_i} y_i, & \tilde{x}_i &= \sqrt{w_i} x_i \\ \tilde{Y} &= W^{1/2} Y, & \tilde{X} &= W^{1/2} X\end{aligned}$$

A regression of $\tilde{Y}$ on $\tilde{X}$:

$$\hat{\beta}_{WLS} = (\tilde{X}' \tilde{X})^{-1} \tilde{X}' \tilde{Y} = (X' W X)^{-1} X' W Y$$

Also used as a solution to heteroskedasticity

- ▶ Relax LSA (2) $(X_i, Y_i), i = 1, \dots, n$, are i.i.d.
- ▶ Relax LSA (4) $Var(u_i) = \sigma^2$ (homoskedasticity)

Why? We are minimizing weighted sum of squared residuals:

$$\sum_{i=1}^N w_i (y_i - \hat{y}_i)^2 = \sum_{i=1}^N w_i u_i^2$$

Suppose we have heteroskedasticity so that $Var(\varepsilon_i) = \sigma_i^2$ and $w_i \propto \frac{1}{\sigma_i^2}$.

In this setting WLS is **BLUE**.

Why does anyone ever run OLS instead of WLS?

- ▶ Problem is that σ_i^2 is unknown before we run our regression.
- ▶ We can estimate $\hat{\sigma}_i^2$.

This procedure is known as Iteratively Re-weighted Least Squares **IRLS**

1. Initialize weights to identity matrix: $W = I$
2. Regress Y on X with weights W
3. Obtain $\hat{u}_i$.
4. Update W with $w_{ii} = \frac{1}{\hat{u}_i^2}$
5. Repeat until parameter estimates don't change

There is no reason to require that W be diagonal. This gives us **Generalized Least Squares**

$$\hat{\beta}_{GLS} = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'\tilde{Y} = (X'\Omega X)^{-1}\Omega'WY$$

The idea is to use the **inverse covariance matrix** of residuals. But this is high dimensional ($N \times N$) and estimating it is harder than our original problem!

Feasible Generalized Least Squares **FGLS**:

1. Initialize weights to identity matrix: $\hat{\Omega} = I$
2. Regress Y on X with weighting matrix $\hat{\Omega}$
3. Obtain $\hat{u}_i$.
4. Construct $\mathbb{E}[u_i^2 \mid X, Z]$ via (nonlinear) regression: $\exp[\gamma_0 + \gamma_1 x_i + \gamma_2 z_i]$.
5. Update $\hat{\Omega}$ with $\mathbb{E}[u_i^2 \mid X, Z]$
6. Repeat until parameter estimates don't change

Thanks!
